

Lab 1 - Task 1

```
import pandas as pd
import numpy as np

aqi = pd.read_csv("city_day.csv")
crop = pd.read_csv("crop_production.csv")

print("="*50)
print("AQI DATASET PROFILE")
print("="*50)

print("Shape :", aqi.shape)
print("\nColumns")
print(aqi.columns)

print("\nData Types")
print(aqi.dtypes)

print("\nMissing Values")
print(aqi.isnull().sum())

print("\nDuplicate Rows")
print(aqi.duplicated().sum())

print("\nStatistical Summary")
print(aqi.describe())

print("\n\n"+"="*50)
print("CROP DATASET PROFILE")
print("="*50)

print("Shape :", crop.shape)

print("\nColumns")
print(crop.columns)

print("\nData Types")
print(crop.dtypes)

print("\nMissing Values")
print(crop.isnull().sum())

print("\nDuplicate Rows")
print(crop.duplicated().sum())

print("\nStatistical Summary")
print(crop.describe())

=====
AQI DATASET PROFILE
=====
Shape : (29531, 16)

Columns
Index(['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2',
      'O3', 'Benzene', 'Toluene', 'Xylene', 'AQI', 'AQI_Bucket'],
      dtype='object')

Data Types
City          object
```

```

Date          object
PM2.5         float64
PM10          float64
NO            float64
NO2           float64
NOx           float64
NH3           float64
CO            float64
SO2           float64
O3            float64
Benzene       float64
Toluene       float64
Xylene        float64
AQI           float64
AQI_Bucket    object
dtype: object

```

Missing Values

```

City          0
Date          0
PM2.5         4598
PM10          11140
NO            3582
NO2           3585
NOx           4185
NH3           10328
CO            2059
SO2           3854
O3            4022
Benzene       5623
Toluene       8041
Xylene        18109
AQI           4681
AQI_Bucket    4681
dtype: int64

```

Duplicate Rows

```
0
```

Statistical Summary

	PM2.5	PM10	NO	NO2	NOx \
count	24933.000000	18391.000000	25949.000000	25946.000000	25346.000000
mean	67.450578	118.127103	17.574730	28.560659	32.309123
std	64.661449	90.605110	22.785846	24.474746	31.646011
min	0.040000	0.010000	0.020000	0.010000	0.000000
25%	28.820000	56.255000	5.630000	11.750000	12.820000
50%	48.570000	95.680000	9.890000	21.690000	23.520000
75%	80.590000	149.745000	19.950000	37.620000	40.127500
max	949.990000	1000.000000	390.680000	362.210000	467.630000

	NH3	CO	SO2	O3	Benzene \
count	19203.000000	27472.000000	25677.000000	25509.000000	23908.000000
mean	23.483476	2.248598	14.531977	34.491430	3.280840
std	25.684275	6.962884	18.133775	21.694928	15.811136
min	0.010000	0.000000	0.010000	0.010000	0.000000
25%	8.580000	0.510000	5.670000	18.860000	0.120000
50%	15.850000	0.890000	9.160000	30.840000	1.070000
75%	30.020000	1.450000	15.220000	45.570000	3.080000
max	352.890000	175.810000	193.860000	257.730000	455.030000

	Toluene	Xylene	AQI
count	21490.000000	11422.000000	24850.000000

mean	8.700972	3.070128	166.463581
std	19.969164	6.323247	140.696585
min	0.000000	0.000000	13.000000
25%	0.600000	0.140000	81.000000
50%	2.970000	0.980000	118.000000
75%	9.150000	3.350000	208.000000
max	454.850000	170.370000	2049.000000

CROP DATASET PROFILE

Shape : (246091, 7)

Columns

```
Index(['State_Name', 'District_Name', 'Crop_Year', 'Season', 'Crop', 'Area',
      'Production'],
      dtype='object')
```

Data Types

```
State_Name      object
District_Name   object
Crop_Year       int64
Season          object
Crop            object
Area            float64
Production       float64
dtype: object
```

Missing Values

```
State_Name      0
District_Name   0
Crop_Year       0
Season          0
Crop            0
Area            0
Production      3730
dtype: int64
```

Duplicate Rows

0

Statistical Summary

	Crop_Year	Area	Production
count	246091.000000	2.460910e+05	2.423610e+05
mean	2005.643018	1.200282e+04	5.825034e+05
std	4.952164	5.052340e+04	1.706581e+07
min	1997.000000	4.000000e-02	0.000000e+00
25%	2002.000000	8.000000e+01	8.800000e+01
50%	2006.000000	5.820000e+02	7.290000e+02
75%	2010.000000	4.392000e+03	7.023000e+03
max	2015.000000	8.580100e+06	1.250800e+09

Lab 1 - Task 2

```
print("Before Cleaning")
print(aqi.isnull().sum())
```

```
# Numeric Columns
```

```
num_cols = aqi.select_dtypes(include=np.number).columns
```

```

for col in num_cols:
    aqi[col] = aqi[col].fillna(aqi[col].median())

# AQI Bucket

aqi["AQI_Bucket"] = aqi["AQI_Bucket"].fillna(
    aqi["AQI_Bucket"].mode()[0]
)

print("\nAfter Cleaning")
print(aqi.isnull().sum())

# Crop Dataset

print("\nCrop Before")
print(crop.isnull().sum())

crop["Production"] = crop["Production"].fillna(
    crop["Production"].median()
)

print("\nCrop After")
print(crop.isnull().sum())

```

Before Cleaning

City	0
Date	0
PM2.5	4598
PM10	11140
NO	3582
NO2	3585
NOx	4185
NH3	10328
CO	2059
S02	3854
O3	4022
Benzene	5623
Toluene	8041
Xylene	18109
AQI	4681
AQI_Bucket	4681
dtype: int64	

After Cleaning

City	0
Date	0
PM2.5	0
PM10	0
NO	0
NO2	0
NOx	0
NH3	0
CO	0
S02	0
O3	0
Benzene	0
Toluene	0
Xylene	0
AQI	0
AQI_Bucket	0

```

dtype: int64

Crop Before
State_Name      0
District_Name   0
Crop_Year       0
Season          0
Crop            0
Area           0
Production      3730
dtype: int64

Crop After
State_Name      0
District_Name   0
Crop_Year       0
Season          0
Crop            0
Area           0
Production      0
dtype: int64

```

Lab 1 - Task 3

```

aqi_before = len(aqi)
crop_before = len(crop)

aqi = aqi.drop_duplicates()
crop = crop.drop_duplicates()

aqi_after = len(aqi)
crop_after = len(crop)

print("AQI Records Removed :", aqi_before-aqi_after)
print("Crop Records Removed :", crop_before-crop_after)

# Standardize text columns

aqi["City"] = aqi["City"].str.strip().str.title()

crop["State_Name"] = (
    crop["State_Name"]
    .str.strip()
    .str.title()
)

crop["District_Name"] = (
    crop["District_Name"]
    .str.strip()
    .str.title()
)

print("Standardization Complete")

AQI Records Removed : 0
Crop Records Removed : 0
Standardization Complete

```

Lab 1 - Task 4

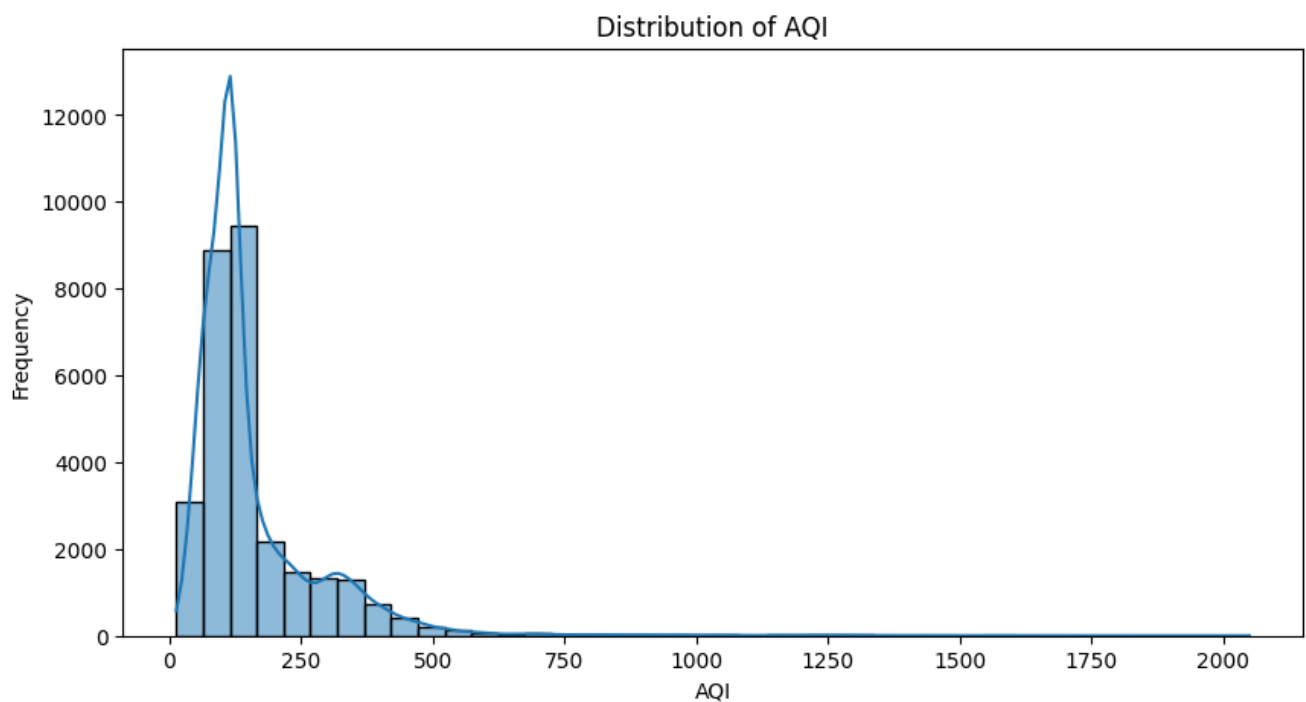
```
import matplotlib.pyplot as plt
import seaborn as sns

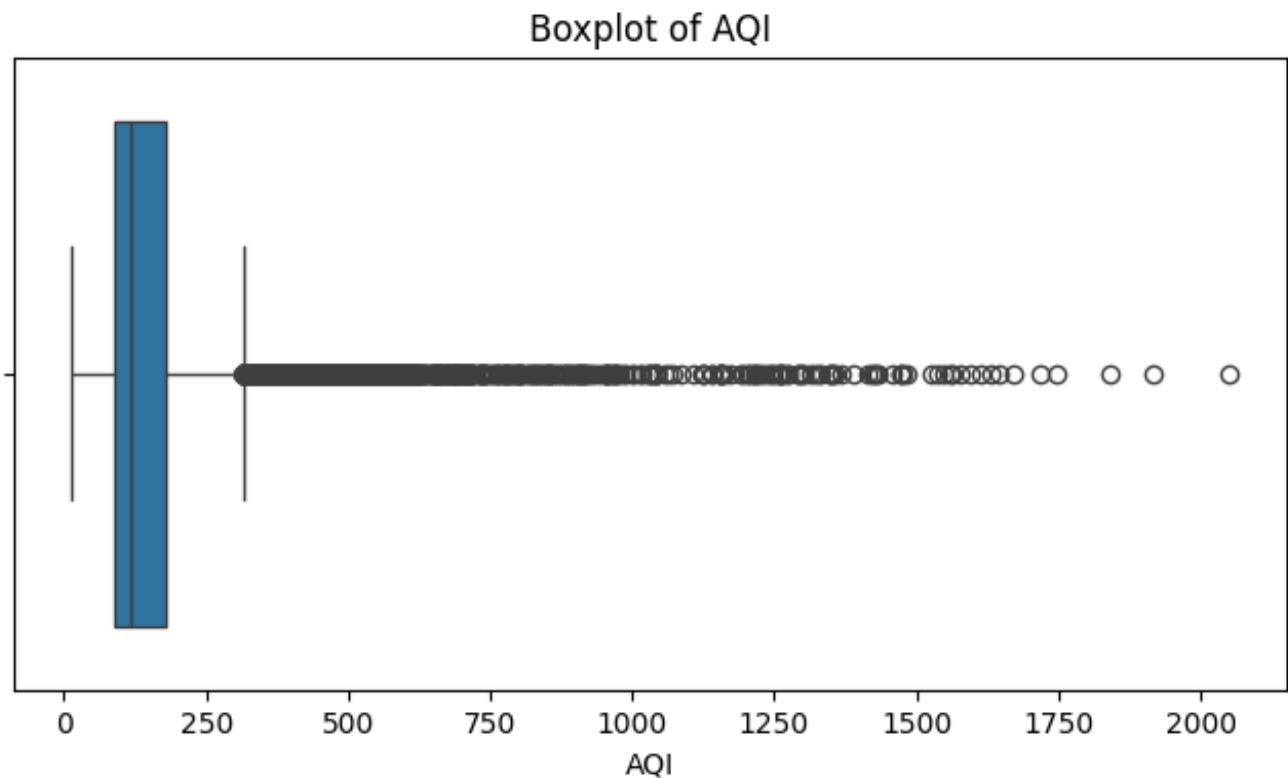
plt.figure(figsize=(10,5))
sns.histplot(aqi["AQI"], bins=40, kde=True)

plt.title("Distribution of AQI")
plt.xlabel("AQI")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(8,4))
sns.boxplot(x=aqi["AQI"])

plt.title("Boxplot of AQI")
plt.show()
```





Lab 1 - Task 5

```
Q1 = aqi["AQI"].quantile(0.25)
Q3 = aqi["AQI"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR
```

```
outliers = aqi[
    (aqi["AQI"] < lower) |
    (aqi["AQI"] > upper)
]
```

```
print("Outliers Found :", len(outliers))
```

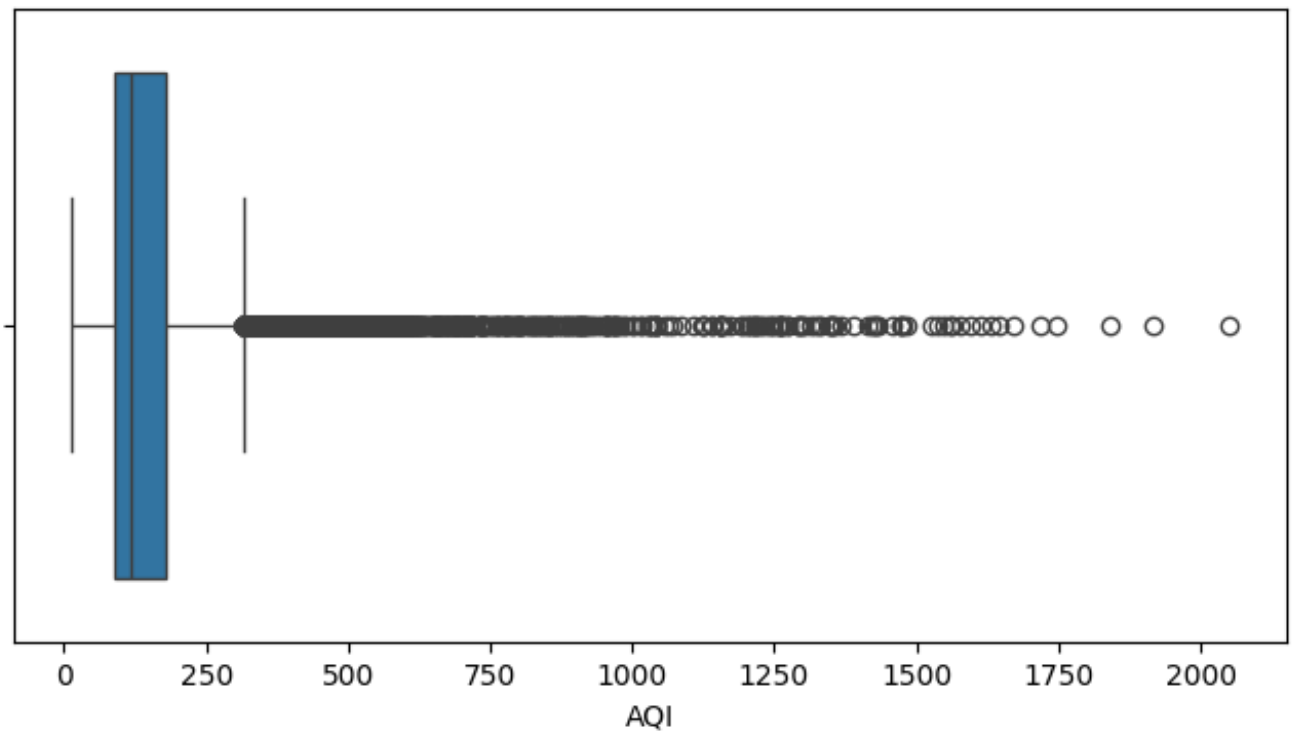
```
plt.figure(figsize=(8,4))
sns.boxplot(x=aqi["AQI"])
plt.title("Before Treatment")
plt.show()
```

```
aqi["AQI"] = np.where(
    aqi["AQI"] > upper,
    upper,
    aqi["AQI"]
)
```

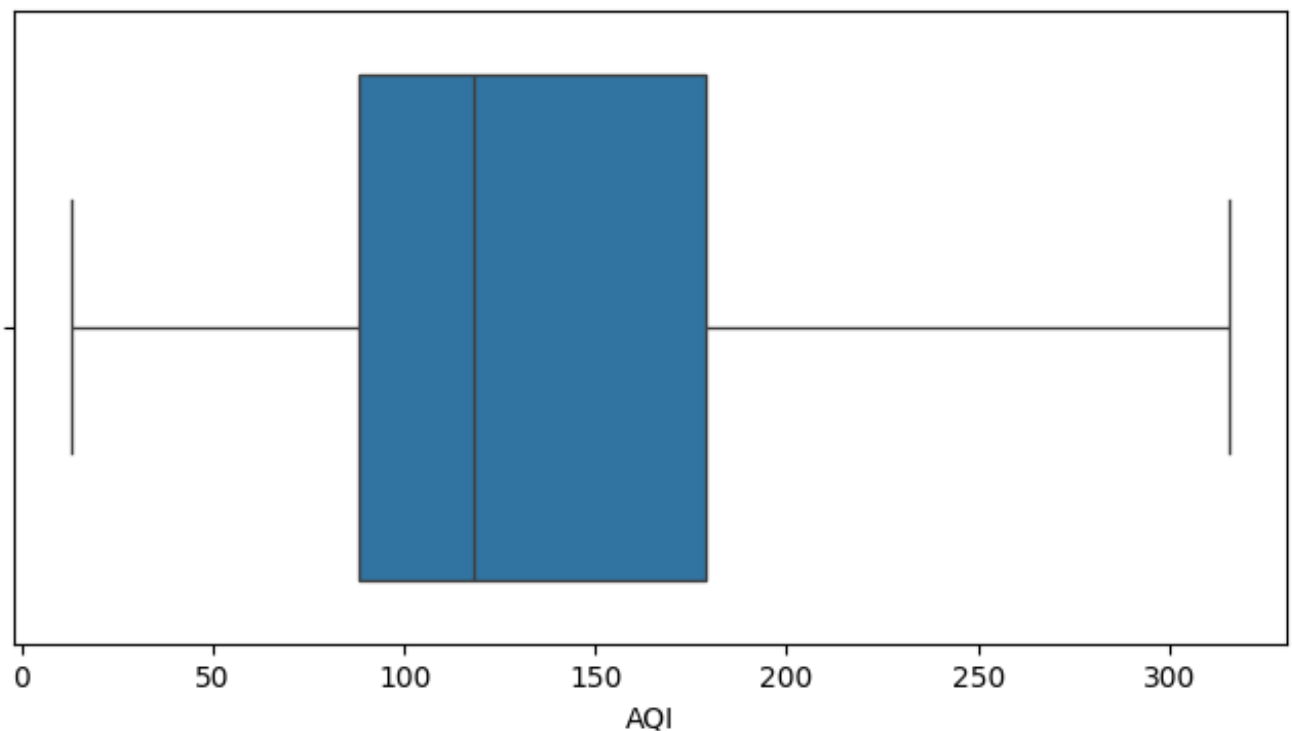
```
plt.figure(figsize=(8,4))
sns.boxplot(x=aqi["AQI"])
plt.title("After Treatment")
plt.show()
```

Outliers Found : 3192

Before Treatment



After Treatment



CONCLUSION

The AQI and crop production datasets were thoroughly inspected to assess their quality and suitability for analysis. Several issues were identified, including missing values, duplicate records, inconsistent text formatting, and extreme AQI readings. Appropriate preprocessing techniques such as median imputation, duplicate removal, text standardization, and outlier treatment were applied to improve data reliability. Visual analysis of AQI revealed that

pollution levels are not uniformly distributed, with most observations concentrated in moderate ranges and a small number of extreme values significantly affecting summary statistics. After cleaning and validation, both datasets became more consistent and suitable for further exploratory analysis and machine learning applications. This preprocessing stage highlights the importance of understanding and preparing raw data before drawing conclusions or building predictive models.